

Network Alignment enhanced via modeling heterogeneity of anchor nodes

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ABSTRACT

Network Alignment (NA), which aims to find the nodes that represent the same entity (i.e., anchor nodes) across different networks, is a fundamental problem in many cross-network researches. Recent advances in network embedding have inspired various auspicious approaches for addressing the NA task, and embedding-based NA technology has become the main research trend. Most embedding-based NA methods follow the consistency assumption explicitly or implicitly, where anchor nodes across different networks tend to have similar local structures/neighbors. However, through the detailed statistical analysis across networks, we observe that anchor nodes have high heterogeneity, i.e., they have different local structures across different networks. Hence, in this paper, we present the formal definition of the heterogeneity of anchor nodes and propose a network alignment framework that combines heterogeneity, which can simultaneously consider the case of heterogeneity and consistency of anchor nodes. In our approach, we propose to use a variational autoencoder to learn node embeddings, and design an effective dual constraint mechanism—Laplacian regularization and heterogeneity constraint to balance the consistency and heterogeneity for network alignment across different networks respectively. Finally, to verify the effectiveness of our proposed method, we conduct extensive experiments on several real-world datasets. Experimental results show that the proposed model achieves better performance than state-of-the-art methods.

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1. Introduction

With the growth of various networks, cross-network analysis attracts growing attention in network science [1–3]. Network Alignment (NA), as one of the most crucial tasks in cross-network analysis, is to find the nodes exist simultaneously in different networks and represent the same natural entity [4–6]. Such nodes are often called as anchor nodes. Network alignment connects different networks by anchor nodes (e.g., find people that active in more than one online social network such as Facebook, Twitter, Weibo, LinkedIn, etc.) to improve related applications, such as online personalized service [7], social link prediction [8–10], recommender system [11–13], protein–protein interaction networks analysis [14], and criminal behavior detection [15].

Early studies mostly utilize node attributes information to calculate the similarity for determining whether two nodes are

anchor nodes in different networks. Due to the privacy policy, node attributes sometimes are unavailability and even untrustworthy. The off-the-shelf network structure is much more reliable in terms of correctness and completeness, and some researchers have begun to introduce network structure into NA. Inspired by network embedding technology [16–18], which has the powerful ability of learning representation of network structures, many embedding-based network alignment approaches have been proposed [19–21]. These methods usually involve two steps: (1) network embedding, which maps the nodes into a low-dimensional space while maintaining the characteristics of the network structure, (2) network alignment, which learns a mapping function to align nodes across different networks.

In the network alignment process, most embedding-based methods hold the **consistency assumption** that **anchor nodes tend to have similar local structures/neighbors across different networks**. Such as people like v_3 and u_4 in Fig. 1(a) maintaining similar local structural features across networks, which meet the assumption of consistency. Nevertheless, the anchor nodes would not follow the consistency assumption absolutely in reality. For

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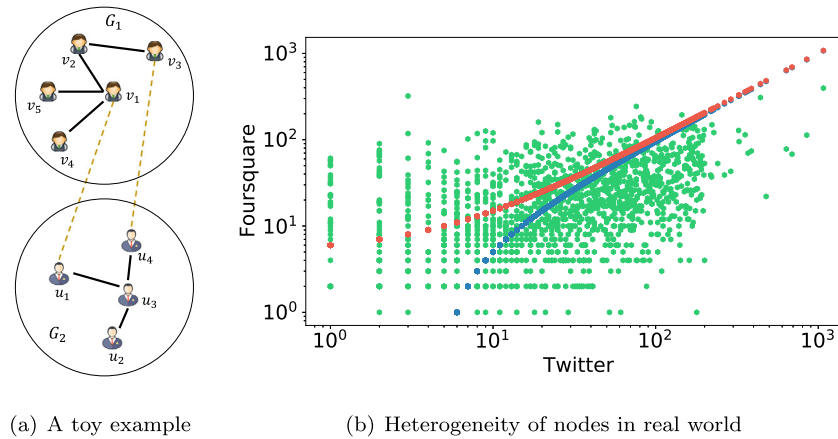


Fig. 1. Illustration of the heterogeneity of anchor nodes. (a) Given two networks G_1 and G_2 , the solid black lines within each network represent the relationships between people, and the yellow dashed line denotes the person across different networks that represent the same natural person. (b) We show anchor nodes degrees in Twitter and Foursquare respectively in double logarithmic coordinate. The red nodes and the blue nodes represent the degree of the anchor node in Twitter plus or minus 5, respectively.

example, a user maintains a relationship with more people in one social network (e.g., Facebook) while remaining in touch with a small number of people in another network (e.g., LinkedIn). That causes the anchor nodes to have different local structural features among the networks. As shown in Fig. 1(a), v_1 and u_1 are a pair of anchor nodes, it is obvious that they have different degrees across networks, which could not meet the assumption of consistency. Hence, we define this phenomenon as **heterogeneity**, i.e., **anchor nodes have different local structures across networks**. Heterogeneity and consistency are complementary states of anchor nodes, heterogeneity is also a non-negligible state of anchor nodes in network alignment.

We further analyze anchor node heterogeneity in two real-world social networks from Twitter and Foursquare [22] in Fig. 1(b). We define the degree discrepancy as d (i.e., $d = d_1 - d_2$, d_1 and d_2 are the degrees of corresponding anchor nodes in Foursquare and Twitter respectively). We find that about 75% of anchor node pairs with $d > 5$ (nodes outside the range enclosed by the red and blue nodes) may violate consistency and reflect the heterogeneity of anchor nodes across networks. However, most embedding-based NA methods only consider the consistency of anchor nodes and ignore its heterogeneity, result in poor alignment results when anchor nodes are more heterogeneous across networks.

Hence, in this paper we propose a network alignment method based on a variational autoencoder, which introduces anchor nodes heterogeneity while considering consistency. Firstly, we use a variational autoencoder to capture the global network structure in our network embedding module, which preserves the topology structures of nodes in a single network. Secondly, the core of our model is that we design a dual constraint mechanism including *Laplacian regularization* and *heterogeneity constraint*, where Laplacian regularization to maintain the consistency assumption, and the heterogeneity constraint to allow the corresponding anchor nodes to maintain heterogeneity in different networks. Laplacian regularization on known anchor nodes across different networks to make the distance between anchor node pairs closer. This constraint also makes the distance between two nodes from different networks closer, if they have common anchor nodes as neighbors. We also utilize the degree discrepancy across nodes and heterogeneity coefficient across networks to depict the heterogeneity, then let the node prefer to align a node that with a large degree discrepancy. Thirdly, we use a decoder to preserve the structure of the network by reconstructing the adjacency matrix.

Through the above modules, we could learn node embeddings that combines the heterogeneity of anchor nodes across networks. Finally, we propose an end-to-end unified framework which integrates **Degree Heterogeneity** and the node embedding model for **Network Alignment (DHNA)**. The main contributions of this paper are summarized as follows:

- Through statistical analysis of several standard datasets, we found that the anchor nodes in different networks have heterogeneity (i.e., the degrees of anchor nodes in different networks are different), and this characteristic has a certain generality. We apply this conclusion to the network alignment task for verifying its usefulness.
- To capture the relationship between anchor nodes with heterogeneity in different networks, we utilize Laplacian regularization and degree discrepancy of nodes as constraints to make a trade-off between the consistency and heterogeneity in network alignment.
- We propose an end-to-end framework DHNA based on a variational autoencoder to solve the network alignment problem. Extensive experiments on real-world datasets have been conducted, and the results show that our method significantly outperforms existing network alignment methods.

We organize our work as follows. Section 2 presents the related works of network embedding and network alignment. We give a formal definition to measure the heterogeneity across node pairs and network pairs respectively in Section 3. Section 4 formally describes our proposed method in detail. We validate the effectiveness of our approach by analyzing extensive experiments in Section 5. A case study is given in Section 6. Section 7 concludes this paper.

2. Related work

We mainly introduce network embedding methods and network alignment methods in this section.

2.1. Network embedding

The methods of network embedding can be broadly categorized into two categories. The first category is single network embedding, such as random walk based [23–25] and graph neural network based [16,26,27]. Besides, there exists network embedding for heterogeneous networks [28–30]. The heterogeneity of heterogeneous networks is due to the multi-types of nodes and

edges, which are different from the heterogeneity proposed in this paper, so these methods are not suitable for direct application in network alignment with anchor nodes heterogeneity.

The second category is multiplex network embedding [31–34]. For example, UDA-GCN [34] simultaneously introduces a local graph convolutional network and a global graph convolutional network in feature aggregation and designs an attention mechanism to weigh the importance between layers. A multiplex network is a network that contains multiple systems of the same set of nodes and there exists various types of relationships among nodes [35]. Network alignment assumes that each network has its own nodes set, and not all nodes have a corresponding node across networks [36]. Therefore, it is ineffective to directly use multiplex network embedding methods to solve the network alignment task.

2.2. Network Alignment

Network alignment is a fundamental task for many fields such as computer vision, chemistry, and pattern recognition. In recent years, extensive researches have been conducted, and the purpose is to find corresponding nodes in different networks by using a mapping function to measure the similarity of nodes. The works in the early period are mainly heuristic. For example, IsoRank [37] aims to find the global alignment of PPI networks, assuming that two proteins would be aligned if their neighbors were well matched. Tereza et al. [38] used user ids and their tags in different social networks to identify the same user, where the more tag frequencies of the corresponding profiles, the two users to be more similar.

With the blooming of network embedding in recent, many network alignment methods based on embeddings have been proposed [19–21,39]. For example, REGAL [40] follows the assumption that aligned nodes have similar structural connectivity or degrees, and define the structural identity of a node by combining its k -hop neighbors' degrees. Then calculating the cross-network nodes similarity relying on the comparable notions of structural and attribute identity. IONE [19] embeds the follower–followee relationships between users in a social network, and preserves the proximity of users with similar followers/followees in the embedded space.

Most existing methods address the network alignment problem based on the consistency assumption, i.e., if two nodes have similar common anchor nodes as neighbors/local structures, they have a high probability to be anchor nodes. However, the anchor nodes across different networks are frequently inconsistent on the network structure in the real world, which is called heterogeneity of anchor nodes and is a critical influence for network alignment results. Hence we propose DHNA integrating heterogeneity to learn node embeddings for NA, which could significantly improve the performance of network alignment.

3. Preliminary

In this section, we introduce the formalized description of the heterogeneity of anchor nodes at the node level and network level respectively, and the problem definition. A list of major notations involved in this paper is summarized in Table 1.

3.1. Problem formulation

Let $G = \{G^1, G^2, \dots, G^l\}$ denote a set with l networks to be aligned, where $G^o = (V^o, E^o)$, $o \in \{1, 2, \dots, l\}$ represents the o th network. $V^o = \{v_1^o, v_2^o, \dots, v_{n_o}^o\}$ is the set of nodes in the o th network, where $n_o = |V^o|$ is the number of nodes in V^o . $E^o = \{(v_i^o, v_j^o) | i, j \in \{1, 2, \dots, n_o\}\}$ reflects the o th network relationships among nodes.

Table 1

Descriptions of major notations.

Notation	Description
G	The set of networks to be aligned
l	Number of networks in G
G^o, V^o, E^o	A network in G and its nodes set and edges set
V^a, E^a	Anchor nodes set and anchor node pairs set of G
v_i^o, v_j^t	Nodes v_i, v_j in networks G^o, G^t
n_o	Number of nodes in G^o
d_i^o	Degree of node v_i in network G^o
$\mathcal{G}$	Network merged by arbitrary two networks in G
$n_{\mathcal{G}}$	Number of nodes in $\mathcal{G}$
A	The adjacency matrix of $\mathcal{G}$
A^o	The adjacency matrix of G^o
$\tilde{A}$	the symmetrically normalized adjacency matrix of $\mathcal{G}$
$\mathcal{A}$	The adjacency matrix of anchor nodes in $\mathcal{G}$
$\mathcal{D}$	Nodes degree discrepancies matrix between the two networks in $\mathcal{G}$
D	The degree diagonal matrix of $\mathcal{G}$
$r_{(v_i^o, v_j^t)}, r^{ot}$	The heterogeneity at the node level and the network level
U	The matrix of nodes embedding that aggregated 1-hop neighbors
W_0, W_{μ}, W_{σ}	The weight matrices of encoder
m	Dimension of input node attributes
z_i	Embedding of v_i
d	Dimension of z_i
Z	The matrix of nodes embedding of $\mathcal{G}$
μ, σ	The matrix of mean vectors and variance vectors
μ_i, σ_i	Mean vector and variance vector
$\delta(\cdot)$	The non-linear activation function
$p(\cdot), q(\cdot)$	Probability distribution
$\hat{A}$	The reconstructed adjacency matrix
$\hat{A}(i, :)$	The i th row of $\hat{A}$

Definition 1 (Anchor Nodes and Anchor Node Pairs). There exist some nodes in different networks indicating the same real-world entities. They are called anchor nodes and are denoted as V^a . We use anchor node pairs to denote the corresponding relations between anchor nodes. The set of anchor node pairs is defined as $E^a = \bigcup_{1 \leq o < t} E_{ot}^a$. E_{ot}^a is the set of anchor node pairs between G^o and G^t , in which each pair of nodes, i.e., $(v_i^o, v_j^t) \in E_{ot}^a$, represent the same entity in the real world. The anchor node pairs follow the transitivity law defined in [41], i.e., if $(v_i^o, v_j^t) \in E^a$ and $(v_j^t, v_k^s) \in E^a$, then $(v_i^o, v_k^s) \in E^a$.

In this paper, we only consider the node degree as a measure of heterogeneity because it can intuitively reflect how active the node is in a network. Thus, we give the following definition of heterogeneity at the node level and the network level respectively.

Definition 2 (Heterogeneity at the Node Level). For two nodes from G^o and G^t respectively, we use degree discrepancy of node pairs as their heterogeneity:

$$r_{(v_i^o, v_j^t)} = |d_i^o - d_j^t|, \quad (1)$$

where d_i^o and d_j^t are the degree of v_i^o and v_j^t respectively.

Definition 3 (Heterogeneity at the Network Level). Motivated by the degree of relevance and coefficient of congruence in the network [42], we define the heterogeneity coefficient across G^o and G^t as:

$$r^{ot} = \frac{\sum_{(v_i^o, v_j^t) \in E_{ot}^a} d_i^o d_j^t - \frac{(\Theta^{ot})^2}{\Lambda^{ot}}}{\Theta^{ot} - \frac{(\Theta^{ot})^2}{\Lambda^{ot}}}, \quad (2)$$

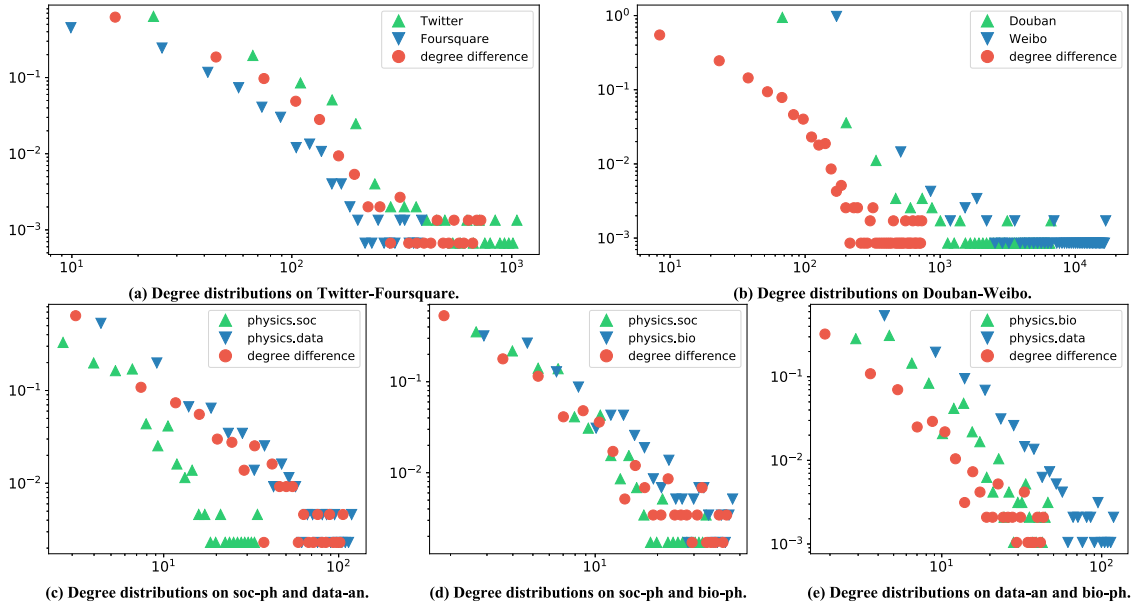


Fig. 2. Anchor nodes degree distribution and its degree discrepancy distribution in real datasets.

where d_i^o and d_j^t are the degree of anchor nodes v_i^o and v_j^t in G^o and G^t , respectively. And Θ^{ot} and Λ^{ot} are defined as follows:

$$\Theta^{ot} = \sum_{(v_i^o, v_j^t) \in E_{ot}^a} \frac{1}{2}((d_i^o)^2 + (d_j^t)^2), \quad (3)$$

$$\Lambda^{ot} = \sum_{(v_i^o, v_j^t) \in E_{ot}^a} (d_i^o + d_j^t). \quad (4)$$

The heterogeneity coefficient $r^{ot} \in (0, 1]$ describes the diversity of anchor nodes' degree distribution in G^o and G^t . If each pair of anchor nodes have the same degrees in G^o and G^t , $r^{ot} = 1$.

As shown in Fig. 2, for the commonly used datasets, we analyze the probability distribution of anchor nodes degrees in different networks and the distribution of degree discrepancy of anchor nodes. It clearly shows that anchor nodes have different local structural features in different networks (probability distributions have only a small part overlapped).

Definition 4 (Network Alignment). Given G , a set of known anchor nodes V^a and anchor edges E^a , the goal of network alignment task is to discover the potential anchor nodes among networks of G , i.e., for a pair of nodes (v_i^o, v_j^t) , where $v_i^o \in V^o$ and $v_j^t \in V^t$, the task is predicting whether a link between them, and it can be formulated as: $f : (G, V^a, E^a) \rightarrow Y, Y \in \{0, 1\}^{n_o \times n_t}$, $o, t \in \{0, 1, \dots, l\}$ and $o \neq t$. $Y_{ij}^{ot} = 1$ if v_i^o and v_j^t were predicted to be a pair of anchor nodes across G^o and G^t , and 0 otherwise. It should be noted that the anchor node pairs satisfy the one-to-one matching constraint, and the networks are partially aligned, i.e., not all the nodes have corresponding nodes in other networks.

4. Proposed method

In this section, we will introduce our proposed method DHNA in detail, which aims to fuse the heterogeneity of the anchor nodes into the network alignment process. Our model is based on the variational autoencoder framework and the overview is shown in Fig. 3.

Taking two networks to be aligned as an example, DHNA mainly contains three parts: (a) an encoder which encodes the nodes in G into a low-dimension embedding space; (b) constraints with nodes degrees discrepancy and known anchor nodes

in G for learned node embeddings, which mainly alleviates the bias caused by heterogeneity of anchor nodes and makes the anchor nodes closer in embedding space; (c) a decoder that reconstructs the network. After getting node embeddings, we learn a mapping function f to predict whether a link exists between any arbitrary pair of nodes. We will introduce them in detail as follows. For G with more than two networks, our model could be used by a pairwise combination of these networks.

4.1. Network encoder

In this section, we adopt a network encoder based on graph convolutional neural network to learn the embeddings of nodes in each network. We merge two networks such that we can train a common encoder to project the two networks into approximately same latent space.

Let $\mathcal{G} = \{V^{\mathcal{G}}, E^{\mathcal{G}}\}$ represents the merged network consisting of arbitrary two networks G^o and G^t in G and anchor node pairs set E_{ot}^a between them as shown in Fig. 3, where $V^{\mathcal{G}} = V^o \cup V^t$, $E^{\mathcal{G}} = E^o \cup E^t \cup E_{ot}^a$. The adjacency matrix of $\mathcal{G}$ can be denoted as $\mathbf{A} = \begin{bmatrix} \mathbf{A}^o & \mathcal{A} \\ \mathcal{A}^T & \mathbf{A}^t \end{bmatrix}$, $\mathbf{A} \in \{0, 1\}^{n_{\mathcal{G}} \times n_{\mathcal{G}}}$, where $n_{\mathcal{G}} = n_o + n_t$ is the number of nodes in $\mathcal{G}$. $\mathbf{A}^o$ is the adjacency matrix of G^o , $\mathbf{A}_{ij}^o = 1$ if $(v_i^o, v_j^o) \in E^o$. $\mathbf{A}^t$ is defined similarly. $\mathcal{A} \in \{0, 1\}^{n_o \times n_t}$ is the adjacency matrix of anchor node pairs across G^o and G^t , $\mathcal{A}_{ij} = 1$ if $(v_i^o, v_j^t) \in E_{ot}^a$, and 0 otherwise. $\mathcal{A}^T$ is the transposition of $\mathcal{A}$. We define the degree discrepancy matrix among nodes from different networks as $\mathcal{D} \in \mathbb{R}^{n_o \times n_t}$, $\mathcal{D}_{ij} = |d_i^o - d_j^t|$.

For the NA task, we are more interested in neighboring nodes than distant ones for predicting a node whether is an anchor node, because the features and connectivity of nearby nodes provide more useful information. Besides, to increase the robust of the learned embedding, we learn the node distribution rather than mapping the node into a point in the embedding space. A random sample $\mathbf{z}$ is taken from the distribution rather than generated from the encoder directly.

Therefore, we use the neighboring relations (i.e., 1-hop neighbors and 2-hop neighbors of a node) to generate the distribution of a node. First, we adopt a graph convolution to aggregate the 1-hop neighbors' information and produce a convolved signal matrix $\mathbf{U} \in \mathbb{R}^{n_{\mathcal{G}} \times F}$:

$$\mathbf{U} = \text{ReLU}(\tilde{\mathbf{A}}\mathbf{X}\mathbf{W}_0), \quad (5)$$

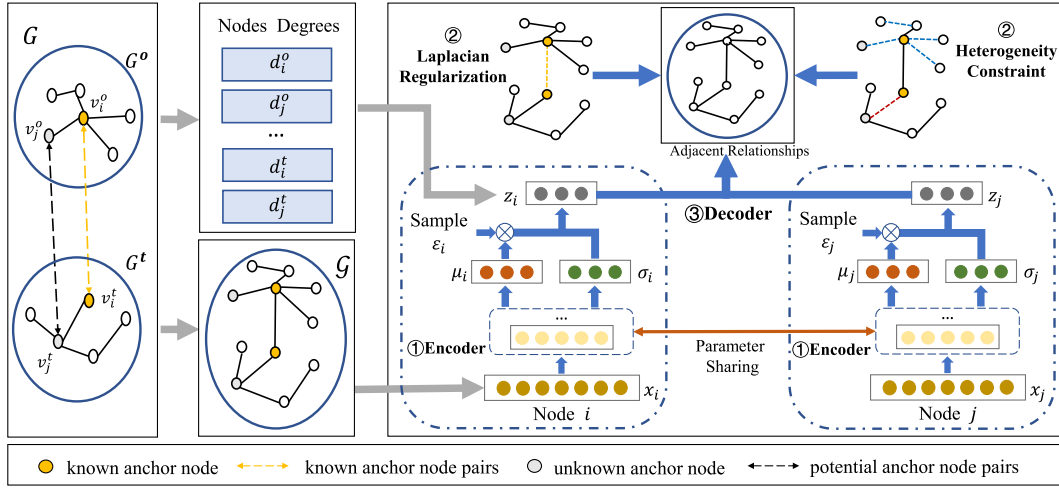


Fig. 3. The framework of DHNA. For arbitrary two networks in G , we merge them as a new network $\mathcal{G}$ and calculate nodes degrees. We firstly obtain the node embeddings by the encoder. Then we use Laplacian regularization on known anchor nodes and heterogeneity constraint to balance the consistency and heterogeneity. Finally, we preserve the node local structure with the decoder.

where $\tilde{\mathbf{A}} = \mathbf{D}^{-\frac{1}{2}}(\mathbf{A} + \mathbf{I})\mathbf{D}^{-\frac{1}{2}}$ is the symmetrically normalized adjacency matrix and $\mathbf{D} \in \mathbb{R}^{n_{\mathcal{G}} \times n_{\mathcal{G}}}$, ($\mathbf{D}_i = \sum_j \mathbf{A}_{ij}$) is a diagonal degree matrix. $\mathbf{I}$ is the identity matrix. $\mathbf{X} \in \mathbb{R}^{n_{\mathcal{G}} \times m}$ is the node attributes matrix of $\mathcal{G}$, where each row is a m -dimensional feature vector. For network without attributes, we let $\mathbf{X} = \mathbf{I}$ as the model input. $\mathbf{W}_0 \in \mathbb{R}^{m \times F}$ is a matrix of filter parameters.

Then we learn nodes distribution based on $\mathbf{U}$. Among all possible probability distributions with the same variance, the normal distribution has the largest uncertainty in real numbers. Therefore, the normal distribution can be considered to be the distribution with the least amount of prior knowledge added to the model, and we assume nodes in $\mathcal{G}$ follow a normal distribution. The normal distribution parameters can be learned by aggregating the nodes 2-hop neighbors' information :

$$\mu = \tilde{\mathbf{A}}\mathbf{U}\mathbf{W}_{\mu}, \quad (6)$$

$$\log \sigma = \tilde{\mathbf{A}}\mathbf{U}\mathbf{W}_{\sigma}, \quad (7)$$

where $\mathbf{W}_{\mu} \in \mathbb{R}^{F \times d}$ and $\mathbf{W}_{\sigma} \in \mathbb{R}^{F \times d}$ are weight matrices with d dimensionality, μ and $\log \sigma$ are the matrix of mean and variance vectors respectively. Embeddings $\mathbf{z}_i$ of v_i are drawn from this distribution:

$$q(\mathbf{z}_i | \mathbf{X}, \mathbf{A}) = \mathcal{N}(\mathbf{z}_i | \mu_i, \text{diag}(\sigma_i^2)). \quad (8)$$

Thus, the nodes learn the information of their neighbors within 2-hop and maintain their local structure in their respective networks.

4.2. Dual constraints for node embeddings

The node embeddings $\mathbf{z}_i$ learned in Section 4.1 only contain the information of its 2-hop neighbors and local structure, and cannot be used for alignment directly because of lacking cross-network information. Therefore, we use known anchor node pairs as a bridge to transfer information across networks. The consistency assumption used in most methods makes potential anchor node pairs tend to have similar local structures and ignore the dissimilar situation. In this section, we propose to design dual constraints to balance the consistency and heterogeneity of anchor nodes in the NA task, including a Laplacian regularization and a heterogeneity constraint.

Laplacian regularization. For learned node representation, we need the anchor nodes across G^o and G^t to have similar

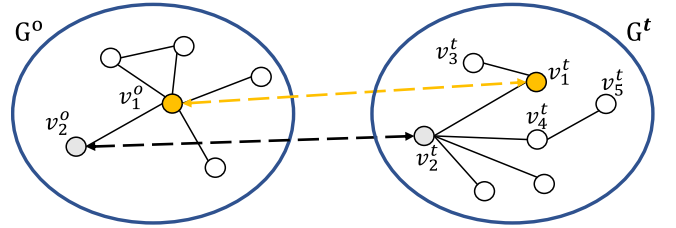


Fig. 4. An example of dual constraints. The Laplacian regularization meets the consistency assumption (i.e., anchor nodes have similar neighbors). The heterogeneity constraint meets the heterogeneity (i.e., anchor nodes have dissimilar local structures).

embeddings. Therefore, we fuse the prior knowledge, i.e., known anchor nodes, to constrain the node embeddings:

$$\begin{aligned} \mathcal{L}_a &= \sum_{(v_i, v_j) \in E_{ot}^a} \|\mathbf{z}_i - \mathbf{z}_j\|_2^2 \mathcal{A}_{ij} \\ &= \text{tr}(\mathbf{Z}^T \mathbf{A} \mathbf{Z}). \end{aligned} \quad (9)$$

Because of the oscillation of $\mathcal{L}_a$, i.e., its values range between positive and negative alternately, we use $\mathcal{L}_a = \text{tr}(\delta(\mathbf{Z}^T) \mathbf{A} \delta(\mathbf{Z}))$ in practice training, where $\delta(\cdot)$ is the sigmoid function. By utilizing the above Laplacian regularization, the embeddings of anchor nodes tend to be the same in theory. As shown in Fig. 4, through the Laplacian regularization, embeddings of anchor nodes v_1^o and v_1^t could be the same. Therefore, the embeddings of nodes v_2^o and v_2^t , v_2^o and v_3^t have a high probability of being the same compared with v_2^o and v_5^t , because they have the same anchor nodes as a neighbor. If the degrees of v_2^o and v_3^t are more similar, then the anchor node neighbors they have in common will have more influence on them. That is, the Laplacian regularization makes nodes with similar degrees more likely to be aligned. This satisfies the consistency assumption followed in most methods.

Heterogeneity constraint. For the heterogeneity, we use the node level heterogeneity, i.e., $\mathcal{D}$, to make the embeddings of potential anchor nodes that have the larger degree discrepancy across different networks closer:

$$\mathcal{L}_h = \sum_i^{n_o} \sum_j^{n_t} \|\mathbf{z}_i^o - \mathbf{z}_j^t\|_2^2 \mathcal{D}_{ij}. \quad (10)$$

Thus, the degree discrepancy will make the embeddings of node pairs with a large degree discrepancy closer. As shown in Fig. 4, v_1^o and v_1^t are anchor node pairs between G^o and G^t . v_2^o is the node to be aligned. Through the Laplacian regularization, v_2^o is closer to v_2^t and v_3^t in the embedding space compared with other nodes. Additionally, the heterogeneity constraint lets v_2^o tend to align with v_2^t because their degree discrepancy is larger compared with the degree discrepancy of v_2^o and v_3^t . It counterbalances a natural effect whereby aligning nodes would have similar degrees, and is a useful complement to other scenarios where anchor nodes satisfy except the consistency assumption.

4.3. Reconstruction decoder

For obtaining nodes representation, the decoder is defined by an inner product between $\mathbf{Z}$. The output of the decoder is a reconstructed adjacency matrix $\hat{\mathbf{A}}$:

$$\hat{\mathbf{A}} = \delta(\mathbf{Z}\mathbf{Z}^T), \quad (11)$$

where $\delta(\cdot)$ is the sigmoid function. Then we define the cross-entropy loss between $\hat{\mathbf{A}}$ and the input adjacency matrix $\mathbf{A}$ when training as follows:

$$\mathcal{L}_{ce} = - \sum_i^{n_o} \sum_j^{n_t} (\mathbf{A}_{ij} + \mathbf{I}_{ij}) \ln \hat{\mathbf{A}}_{ij}. \quad (12)$$

In order to make the result of the reconstruction matrix better, i.e., $\hat{\mathbf{A}}$ is as similar to $(\mathbf{A} + \mathbf{I})$ as possible, $\mathcal{L}_{ce}$ will try to make the variance $\log \sigma$ be 0. Therefore, we let $q(\mathbf{Z} | \mathbf{X}, \mathbf{A}) = \prod_{i=1}^{n_g} q(\mathbf{z}_i | \mathbf{X}, \mathbf{A})$ closer to the standard normal distribution to prevent the variance from becoming all 0:

$$\mathcal{L}_{KL} = KL[q(\mathbf{Z} | \mathbf{X}, \mathbf{A}) \parallel p(\mathbf{Z})], \quad (13)$$

where $KL[q(\cdot) \parallel p(\cdot)]$ is the Kullback–Leibler divergence between $q(\cdot)$ and $p(\cdot)$, $p(\mathbf{Z}) = \prod_i p(\mathbf{z}_i) = \prod_i \mathcal{N}(\mathbf{z}_i | \mathbf{0}, \mathbf{I})$ is a standard normal distribution prior.

The proposed DHNA is jointly trained with the above losses, Laplacian regularization and heterogeneity constraint, the full loss functions are:

$$\mathcal{L} = \mathcal{L}_{ce} + \mathcal{L}_{KL} + \lambda \mathcal{L}_a + \gamma \mathcal{L}_h, \quad (14)$$

where λ and γ are the hyper-parameters that trade-off the consistency and the heterogeneity in network alignment.

Proposition 1. *The optimization object of our model is an upper bound of the distance between two arbitrary nodes from different networks.*

Proof. To better illustrate our analysis, we utilize the example in Fig. 4, and define the distance metric as $d(\cdot, \cdot)$. Taking the loss function of DHNA into consideration, $\mathcal{L}_{ce}$ and $\mathcal{L}_{KL}$ minimize the distance between neighbors in the same network, such as $d(v_1^o, v_2^o)$ and $d(v_1^t, v_2^t)$. $\mathcal{L}_a$ minimizes distance of anchor node pairs, such as $d(v_1^o, v_1^t)$. Therefore, if (v_2^o, v_2^t) is the potential anchor node pairs, it satisfies: $d(v_2^o, v_2^t) \leq d(v_2^o, v_1^o) + d(v_1^o, v_1^t) + d(v_1^t, v_2^t)$.

4.4. Network alignment based on embedding

After acquiring the final node representations of $\mathcal{G}$, we find the potential anchor node pairs based on the reconstructed adjacency matrix $\hat{\mathbf{A}}$. For a given node $v_i \in G^o$, we need to find the most similar node $v_j \in G^t$ for it, i.e., v_i and v_j have the largest similarity of $\hat{\mathbf{A}}(i, :)$. If it is not that strict, we can return a corresponding node list $list_\alpha = \{v_j\}$, $v_j \in G^t$, i.e., the similarity of node pairs (v_i, v_j) ranks in the top α of $\hat{\mathbf{A}}(i, :)$.

More generally, we can define a mapping function f as introduced in Section 3.1 to predict if a pair of nodes (v_i, v_j) , $v_i \in G^o$, $v_j \in G^t$ are anchor node pairs:

$$y_{ij} = \begin{cases} 1, & \text{if } \hat{\mathbf{A}}_{ij} \geq p \\ 0, & \text{otherwise} \end{cases} \quad (15)$$

where p is the threshold that depends on the acceptability of candidate results by users.

4.5. Time complexity

There are three steps to analyze: **Node Embedding**: the time complexity for calculating $\tilde{\mathbf{A}}$ is $O(|E^g|)$. Therefore, the time complexity of learning μ and $\log \sigma$ is $O(2(|E^g| d + (n_o + n_t)d^2)) = O(|E^g| d + (n_o + n_t)d^2)$. **Dual Constraints**: the time complexity of Laplacian regularization and heterogeneity constraint are $O(n_a d)$ and $O(n_o n_t d)$ respectively, where n_a is the number of anchor nodes and $n_a \ll n_o n_t$. Therefore, the time complexity of dual constraints is $O(n_o n_t d)$.

Decoder: the time complexity of decoder is $O(n_o n_t d)$.

Therefore, the total time complexity is $O((n_o + n_t)d^2 + (|E^g| + n_o n_t)d)$.

5. Experiment

In this section, we conduct extensive experiments over five real-world datasets to evaluate the performance of our proposed model DHNA. We will first introduce the datasets and the baseline algorithms, then present the experiment settings. Afterward, we discuss performance evaluation and analysis of model effectiveness.

5.1. Datasets

In our experiments, we select five real-world network alignment datasets, including two social-network datasets and three academic-network datasets. The two social-network datasets are **Twitter-Foursquare** [22], which are collected from Foursquare and Twitter where the ground truth of anchor nodes is provided in the Foursquare profile; **Douban-Weibo** [7], which are the Chinese most prominent microblogging and movie rating sites, and the alignment between them is revealed explicitly in the Douban user profile. The remaining three datasets are a subset of the **ArXiv dataset** [43]. ArXiv dataset contains layers corresponding to different arXiv categories. We randomly select three layers for our experiments. Table 2 lists the statistics of these datasets in detail.

5.2. Baseline methods

We choose eight competitive embedding-based NA methods to validate the effectiveness of the proposed DHNA. These methods and our method only use network structure. The comparison methods are listed as follows:

- **IONE** [19]: It embeds the relationships of follower–followee between nodes in the respective networks, and known anchor nodes are used to align the two networks.
- **PALE** [20]: It employs network embedding to capture the structural regularities on different networks separately and further learns a mapping function to predict anchor nodes with observed anchor links.
- **REGAL** [40]: It exploits the low-rank approximation to factorize the node similarities captured through structural and attribute information (if have).

Table 2

Statistics of the datasets. Heterogeneity coefficient indicates the heterogeneity caused by anchor nodes across networks.

Networks	#Nodes	#Relations	#Anchor links	Average node degree	Heterogeneity coefficient
Twitter	5,120	164,919	1,609	32.211	0.553
Foursquare	5,313	76,972		14.487	
Douban	15,151	49,110	1,170	3.226	0.121
Weibo	28,646	110,982		3.858	
physics.soc-ph	1,594	3,019	434	1.894	0.361
physics.data-an	5,465	14,485		2.651	
physics.soc-ph	1,108	3,019	583	1.894	0.484
physics.bio-ph	1,905	6,097		2.062	
physics.data-an	3,583	14,485	958	2.651	0.493
physics.bio-ph	1,905	6,097		2.062	

- **CrossMNA** [39]: It performs node embedding and anchor nodes alignment by introducing an intra-vector and an inter-vector. Inter-vector reflects the common features of the anchor nodes and intra-vector preserves the specific structural feature for a node in its selected network.
- **MGCN** [21]: It performs convolutions on simple graphs and hypergraphs to learn network embeddings respectively, and reconciles both of them into the same latent space by known anchor node pairs.
- **HALF** [44]: It maps the structural features of different networks into a Hamming space of binary codes to preserve the similarity between user identities.
- **LSNA** [45]: It performs the network internal aggregation to model the potential correlation between identities respectively in single networks, then introduces cross-network convolution to achieve information aggregation between two networks.
- **ORIGIN** [46]: It learns node representations of different networks by multi-graph convolutional networks, learns the point set transformation function and then infers the alignment based on the transformed node representations.

5.3. Experimental settings

In this section, we will introduce our experiment details, including evaluation metrics and parameter settings.

5.3.1. Evaluation metrics

We compare the precision of the alignment for each method. For a given pair of nodes (v_i^o, v_j^t) where $v_i^o \in G^o$ and $v_j^t \in G^t$, we calculate whether v_j^t is one of the top α most similar nodes to v_i^o . We use the following evaluation metric from G^o to G^t followed [39]:

$$P^{(o,t)}@_\alpha = \frac{|M^{(o,t)}@_\alpha|}{|U^{(o,t)}|}, \quad (16)$$

where $|M^{(o,t)}@_\alpha|$ is the count of the correct alignments from G^o to G^t in top- α choices, and $|U^{(o,t)}|$ is the number of anchor node pairs to be aligned. Therefore, the evaluation metric of G can be defined as:

$$P@_\alpha = \frac{1}{l(l-1)} \sum_{o,y \geq 1, o \neq t} P^{(o,t)}@_\alpha, \quad (17)$$

where l is the number of networks in G . $P@1$ is also called hard matching, and $P@_\alpha$ ($\alpha > 1$) is called soft alignment.

5.3.2. Parameter settings

For network alignment, we randomly divide known anchor nodes into a training set and a test set at a ratio of 3 : 1. We set $\lambda = \gamma = 1$, and the learning rate to 0.001 in our model. We set

the same embedding dimension $d = 200$ for all methods, except for CrossMNA and HALF. For CrossMNA, we set the dimension of inter-vector and intra-vector to 200 and 100, respectively. For HALF, we set the length of binary codes as 32 bits. For PALE, we use LINE [47] to learn node embeddings and adopt an MLP model as a mapping function, where the dimension of the hidden layers is $2 * d$. For REGAL, we set the maximum hop distance $K = 2$. For MGCN, we set $N_{max} = 20,000$, $N_{min} = 1000$ and $N = 200$ in the graph partition and reconciliation step. It splits the large network into several partitions with acceptable size (from N_{min} nodes to N_{max} nodes), N is the number of nodes from the network as shared nodes across all partitions. Other hyperparameters of the baseline methods are the same as they are reported in their papers.

5.4. Performance analysis

The experimental results are presented in Table 3, where $P@_\alpha$ of different methods on various datasets. We can find the following observations.

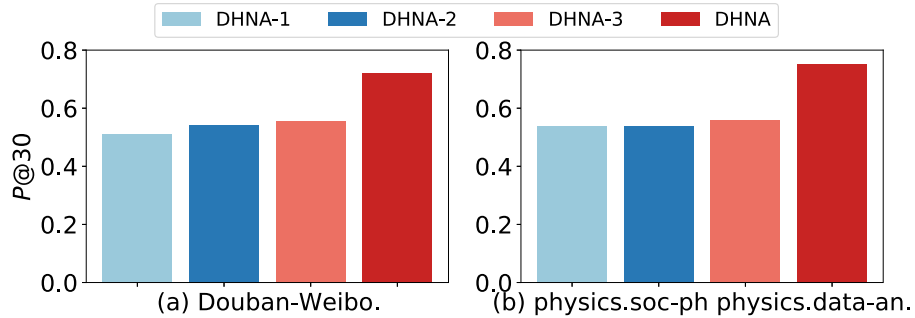
(i) Our method DHNA can outperform most baselines in each dataset. Moreover, compared with all the baselines, our model achieves the minimum 0.83% average improvement in the Douban_Weibo dataset and the maximum 12.90% average improvement on physics.data-an-physics.bio-ph dataset. It demonstrates that the heterogeneity of anchor nodes we discovered in real-world datasets has an important impact on network alignment. Moreover, it can be seen from Table 3 that the results of our model change very little from $P@1$ to $P@30$ compared with most baseline methods, indicating that our method can work well for both hard matching and soft alignment.

(ii) The unsupervised method REGAL shows poor performance on all datasets because the heterogeneity of anchor nodes causes structures of each network no longer to comply with the assumption of consistency, which is the fundamental assumption it holds. It is worth noting that MGCN, CrossMNA, LSNA and ORIGIN also perform well. CrossMNA uses anchor nodes as the intermediate entities that transmit information, but ignores that when the node is heterogeneous, the wrong message will be transmitted. MGCN constructs the hypergraph via each node's 10-hop neighbors, which alleviates the inconsistency to a certain extent. However, considering too wide a range of neighbors does not provide more effective information, and even causes interference. LSNA makes two nodes with similar neighbor nodes more likely to be the corresponding anchor nodes by intra-network convolution and cross-network convolution. This indicates that LSNA is easier to align anchor nodes that satisfy the consistency assumption, which can also be seen from the experimental results (e.g., LSNA performs better in the Twitter-Foursquare dataset than in the Douban-Weibo dataset since more anchor node pairs exist heterogeneity in Douban-Weibo dataset according to heterogeneity coefficient). ORIGIN uses the known anchor node pairs

Table 3

Overall performance comparison on five datasets of network alignment task. The best and second-best results are highlighted in boldface and underlined, respectively.

Twitter-Foursquare									
	IONE	PALE	REGAL	CrossMNA	MGCN	HALF	LSNA	ORIGIN	DHNA
$P@1$	0.210	0.217	0.199	0.270	<u>0.665</u>	0.223	0.588	0.502	0.750
$P@10$	0.360	0.287	0.229	0.362	<u>0.692</u>	0.280	0.661	0.540	0.785
$P@30$	0.580	0.412	0.302	0.650	<u>0.750</u>	0.420	0.727	0.558	0.816
Douban-Weibo									
	IONE	PALE	REGAL	CrossMNA	MGCN	HALF	LSNA	ORIGIN	DHNA
$P@1$	0.299	0.328	0.089	0.286	0.652	0.301	<u>0.655</u>	0.600	0.661
$P@10$	0.326	0.419	0.144	0.379	0.669	0.430	<u>0.672</u>	0.640	0.689
$P@30$	0.539	0.613	0.285	0.727	0.709	0.584	0.692	0.677	<u>0.720</u>
physics.soc-ph-physics.data-an									
	IONE	PALE	REGAL	CrossMNA	MGCN	HALF	LSNA	ORIGIN	DHNA
$P@1$	0.326	0.308	0.035	0.182	0.621	0.298	<u>0.629</u>	0.593	0.692
$P@10$	0.476	0.396	0.042	0.375	0.637	0.388	<u>0.650</u>	0.627	0.717
$P@30$	0.634	0.608	0.085	0.745	0.671	0.599	<u>0.700</u>	0.688	0.752
physics.soc-ph-physics.bio-ph									
	IONE	PALE	REGAL	CrossMNA	MGCN	HALF	LSNA	ORIGIN	DHNA
$P@1$	0.371	0.338	0.075	0.189	<u>0.630</u>	0.314	0.627	0.627	0.700
$P@10$	0.466	0.383	0.049	0.381	0.649	0.390	<u>0.650</u>	0.641	0.726
$P@30$	0.628	0.598	0.093	<u>0.755</u>	0.697	0.590	0.720	0.671	0.764
physics.data-an-physics.bio-ph									
	IONE	PALE	REGAL	CrossMNA	MGCN	HALF	LSNA	ORIGIN	DHNA
$P@1$	0.210	0.216	0.199	0.273	<u>0.637</u>	0.209	0.630	0.621	0.715
$P@10$	0.321	0.287	0.229	0.357	<u>0.650</u>	0.266	<u>0.650</u>	0.629	0.750
$P@30$	0.528	0.412	0.302	0.662	<u>0.723</u>	0.399	0.713	0.675	0.803

**Fig. 5.** Results of ablation study.

learning a transformation function to maximize the overlappings of corresponding anchor nodes. Anchor node pairs that do not satisfy the consistency assumption affect the quality of the learned transformation function.

5.5. Ablation study

In this section, we conduct ablation experiments to evaluate the important modules in our model. Three variants are designed: DHNA-1 that consists of $\mathcal{L}_{ce}$ and $\mathcal{L}_{KL}$; DHNA-2 that consists of $\mathcal{L}_{ce}$, $\mathcal{L}_{KL}$ and $\mathcal{L}_a$; DHNA-3 that consists of $\mathcal{L}_{ce}$, $\mathcal{L}_{KL}$ and $\mathcal{L}_h$.

Fig. 5 presents the results. It can be seen that the Laplacian regularization and heterogeneity constraint we used is helpful compared with DHNA-1. Combined with Table 2, we can find that the heterogeneity coefficient of these two datasets is less than 0.6, which indicates that the anchor nodes exhibit significant heterogeneity in datasets. The smaller the heterogeneity coefficient, the better DHNA-3 results than DHNA-2, which means that when meeting the consistency assumption, the heterogeneity constraint can help nodes with different local structures across networks find their match.

5.6. Parameter sensitivity

There are four key hyperparameters in our method: the dimension d of node embeddings, the anchor nodes ratio (the ratio of known anchor nodes in the training set when learning the node embeddings), and the coefficient λ , γ of $\mathcal{L}_a$ and $\mathcal{L}_h$. We pick up one parameter each time and vary that parameter while fixing the others to check the impact of this parameter with $P@1$ as the evaluation metric. And we set $d = 200$, anchor nodes ratio as 0.75, $\lambda = 1$ and $\gamma = 1$ as default.

From Fig. 6(a), we can observe that as the node dimension d increases, the performance improves; when d is greater than 200, it decreases instead. This is because our model acquires the node embedding by sampling from the learned normal distribution, changes in variance increase with the d increases and will introduce noise during the alignment.

From Fig. 6(b), we can observe that our model achieves the best performance when the anchor nodes ratio equals 0.75. The reason is that when the anchor nodes ratio equals 1, it will cause the overfitting of the model and result in declining the model performance.

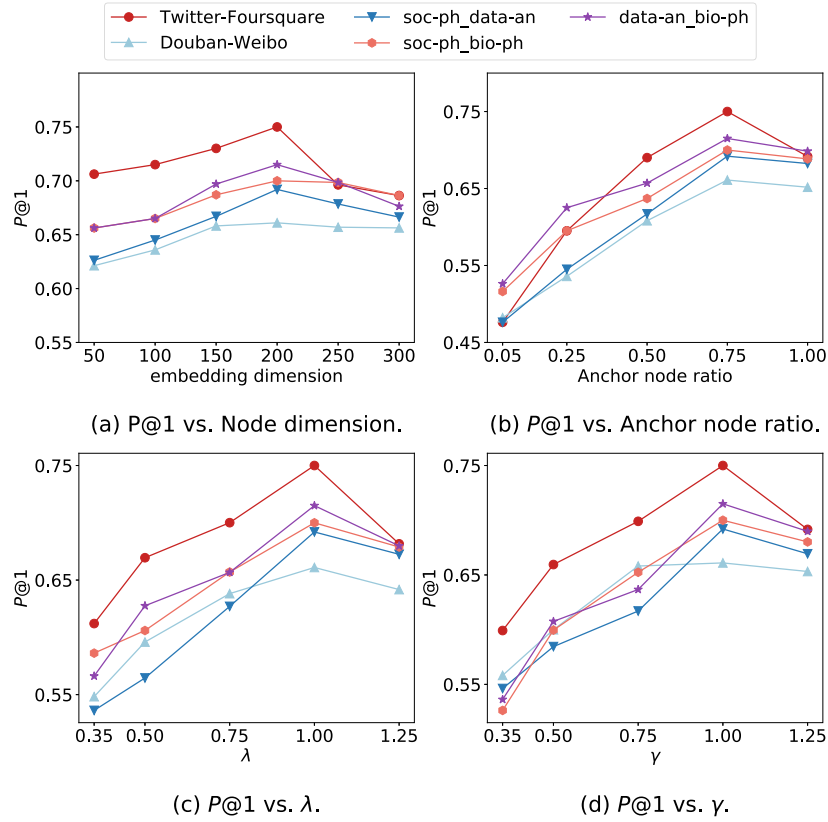


Fig. 6. Parameters study of the proposed DHNA.

From Figs. 6(c) and 6(d), we can observe the effectiveness of the constraint we defined to meet the consistency and heterogeneity of anchor nodes in the NA task, and our model achieves the best performance on $\lambda = 1$ and $\gamma = 1$.

6. Case study

In this section, we conduct a comprehensive case study to validate the effectiveness of DHNA with two sub-networks extracted in the Twitter-Foursquare datasets. As shown in Fig. 7, we display the alignment results of our method and the other eight baseline methods. The blue, green, and red nodes in each sub-graph denote the anchor users in the Twitter test set, anchor users in the Foursquare test set, and mispredicted users in Foursquare (including anchor users and non-anchor users in the Foursquare test set), respectively. The size of a node indicates the degree of that node in this network, i.e., the larger the node is, the greater degree it has. In particular, from Fig. 7 we observe that:

(i) **Nodes with similar degrees tend to be regarded as the anchor nodes across networks by baseline methods.** Generally speaking, the same user should intuitively have similar social relationships in different social networks, which is the foundation of the consistency assumption. Based on this assumption, topologically similar nodes with the help of the clue provided by the anchor node pairs are mapped across different networks, such as node T_{327} and node F_{5447} . Almost all baseline methods can correctly match the anchor nodes with a similar degree across networks, and our method can do the same.

(ii) **Nodes with significant degree discrepancy are difficult to be matched across networks via baseline methods.** In the real world, people tend to interact with different types on different social networks for their purposes in actual communication. For example, people post tweets on Twitter and interact with strangers, and they may interact with their real-life friends in

Foursquare because the geo-location information may expose personal privacy. Thus, when the pair of anchor nodes have a more significant degree discrepancy, such as node T_{53} and node F_{5173} , most baseline methods could not match them correctly, and only our method and MGCN can match correctly. The reason is that such node pairs violate the consistency assumption of these methods. Furthermore, based on consistency assumption, these methods regard node F_{5313} in Foursquare as the corresponding node of T_{53} , where F_{5313} is almost the node with the largest degree in Foursquare.

(iii) **The heterogeneity constraint can be used as an effective supplement to the consistency assumption.** From Fig. 7 we can find that our method can correctly match most anchor nodes with a larger degree discrepancy. However, due to the influence of the extreme degree discrepancy, our method makes erroneous predictions on some anchor node pairs where the degree discrepancy is not so significant, and we will consider how to constrain these extreme values in our subsequent work.

7. Conclusion

This paper studies the problem of network alignment and focuses on analyzing the characteristic of anchor nodes across different networks in the real world. We first find and define the heterogeneity of anchor nodes through the statistical analysis of a large number of standard datasets. Then we design a network alignment method based on the variational autoencoder framework, and integrate the heterogeneity into the encoder module via two well-designed constraints. We evaluate the validity of the proposed method by using five real datasets with a strong heterogeneity coefficient, and extensive experiments showed analytically that the proposed DHNA could acquire significant improvement with most baseline methods. Furthermore, we show a case study on the Twitter-Foursquare dataset, which intuitively

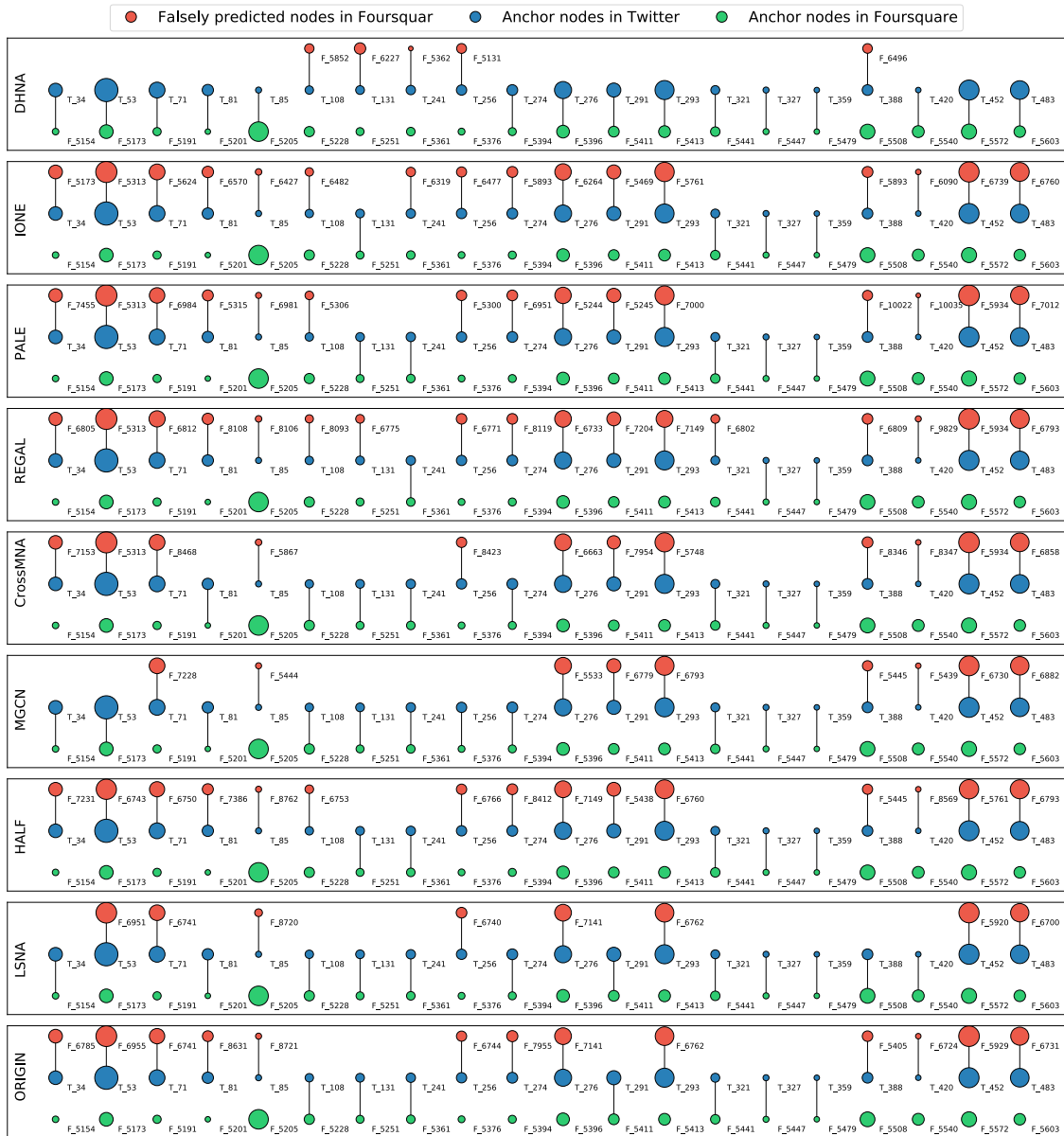


Fig. 7. The partial alignment result visualizations of our method DHNA and other baseline methods. The blue and green nodes denote the correct anchor nodes between Twitter and Foursquare, respectively, and the red nodes represent error anchor nodes in Foursquare. The size of the node indicates the node degree in the corresponding network. It can be seen intuitively from the illustration that our method can realize the alignments of most anchor nodes across networks with a large degree discrepancy, which meets our motivation.

demonstrates the effectiveness of combining heterogeneity as a complement to consistency in network alignment.

CRediT authorship contribution statement

Yinghui Wang: Methodology, Data curation, Writing – original draft. **Qiyao Peng:** Programming. **Wenjun Wang:** Supervision. **Xuan Guo:** Visualization. **Minglai Shao:** Investigation. **Hongtao Liu:** Writing – review & editing. **Wei Liang:** Writing – review & editing. **Lin Pan:** Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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